Reducing false positives in wide-field astronomical streak detection using a lightweight CNN classifier

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Satellites and orbital debris crossing a telescope's field of view leave linear streaks on long-exposure images. These streaks contain information on object orbit, size, and rotation. Streaks found in astronomical archives can thus be exploited for the benefit of Space Situational Awareness (SSA). We build upon an existing deep-learning streak detection pipeline applied to the VLT Survey Telescope (VST) archive. While accurate at localizing streak endpoints, the detector produces a high false positive rate on previously unseen data, preventing reliable large-scale processing. We introduce a lightweight post-detection classifier trained on manually annotated VST cutouts to filter detector outputs. On a withheld test month of archival data, the classifier rejects 96% of false positives while preserving 97% of true streaks, increasing the pipeline F1-score from 0.87 to 0.98. We conduct a systematic study to identify the most effective data representation and model architecture. A lightweight VGG6 convolutional neural network achieves the highest validation performance, while deeper residual models do not provide measurable gains. A multilayer perceptron using only numerical features produced by the detector enables faster inference but at reduced accuracy. The resulting pipeline enables reliable full-archive processing and a more comprehensive analysis of the detected streaks, potentially including uncatalogued debris.

1 Introduction

A streak is produced when a moving object crosses an imager's field of view during a long-exposure observation, appearing as a linear trace in CCD images. Although treated as interference when performing astronomical measurements, streaks contain valuable information. Their geometry and photometric profile allow the estimation of orbit, rotation rate, and size, which are relevant to SSA and debris monitoring [1].

Wide-field surveys accumulate large volumes of long-exposure images over many years. Their archives contain numerous unannotated satellite observations, including uncatalogued objects [1]. Exploiting this information at scale requires reliable automated detection.

Several approaches have been proposed for streak detection. Matched-filtering techniques correlate images with line-shaped templates but are computationally demanding for large datasets [2, 3]. Hough-transform-based methods efficiently identify linear patterns and are widely used in operational pipelines [4, 5], yet they respond to any linear structure and therefore generate false positives. More recent deep-learning approaches integrate detection and classification to improve robustness [6].

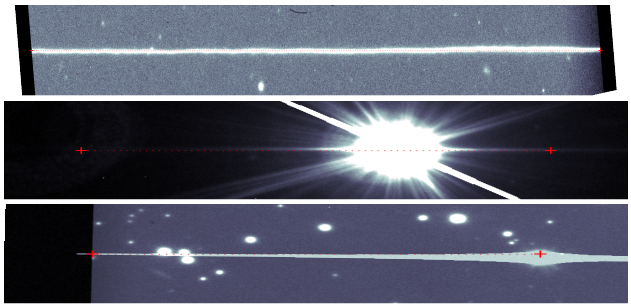


Figure 1: True streak (top) and typical false positives.

Nevertheless, reliable rejection of spurious detections remains essential when processing large survey archives. Specifically, artifacts (diffraction spikes, saturation bleeds, filter fringes) are often misidentified as streaks (Figure 1), preventing large-scale photometric analysis without dedicated filtering.

The detection pipeline on which this work builds provides accurate endpoint localization, but its reliability degrades on previously unseen data. We therefore introduce a lightweight post-detection classifier operating on standardized cutouts around detected candidates. Its objective is to identify false positives while preserving genuine streaks.

The main contributions of this work are:

- The integration of a lightweight convolutional classifier on top of an existing detector, improving the F1-score from 0.87 to 0.98.
- A systematic study of data representation, preprocessing, and classifier architecture, providing insights that could support adaptation to other astronomical image archives.
- The processing of the full 2023 VST/OmegaCAM archive, applying the classifier to 47,680 detector candidates.

By reducing false positives, the approach enables reliable large-scale streak catalogs suitable for downstream

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analyses. The complete pipeline, including source code, annotated datasets, and pre-trained models, is available on GitHub[1] and Zenodo[2].

2 Dataset and Annotation Pipeline

2.1 Data

We use archival data from the VLT Survey Telescope (VST) equipped with OmegaCAM [7], containing long-exposure images where fast-moving objects appear as linear streaks.

The classifier operates on top of an existing HT-LCNN streak detection pipeline applied to the VST archive [8, 9], which outputs candidate streaks defined by their estimated endpoints.

The detector, trained on pre-2023 data, is applied to all VST observations acquired in 2023, which are unseen by the detector.

For each detected candidate, an image cutout centered on the streak is extracted and used as input to the classifier. Candidates from January to March 2023 are used for training, November for validation, and December for testing, resulting in 10,783 training samples, 4,199 validation samples, and 3,370 test samples.

2.2 Annotation

All 18,352 detections used for training, validation, and testing were manually reviewed and labeled as true streaks or false positives.

False positives were further grouped into common artifact categories such as saturation effects, diffraction spikes, and filter fringes. Although the final task is binary classification, this categorization supports error analysis and future extensions.

Annotation was performed using a centralized web-based platform based on the label-studio software package, allowing consistent labeling across the dataset. For each detection, annotators were provided with the cutout, the corresponding CCD image, and its location within the full mosaic to ensure reliable labeling.

3 Methodology

3.1 Classifier Architecture

We adopt a lightweight CNN inspired by the VGG6 architecture introduced in DeepStreaks [6], selected for its simplicity, low parameter count, and proven effectiveness for streak detection on astronomical images. Our implementation is written in PyTorch.

[1] https://github.com/elisabethrachith/odle
[2] https://zenodo.org/records/18097841

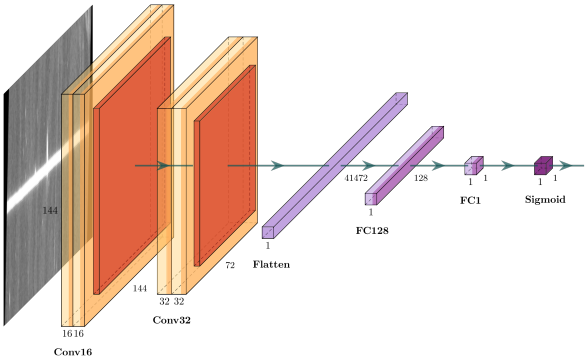


Figure 2: Architecture of the proposed VGG-style classifier.

The architecture is illustrated in Figure 2. The network takes as input a single-channel 144×144 image and consists of two convolutional blocks. Each block contains two 3×3 convolutions (padding 1), each followed by batch normalization and ReLU activation. The first and second blocks use 16 and 32 output channels, respectively. Each block is followed by 2×2 max pooling and dropout (0.25). After the second pooling layer, the network produces 32 feature maps of size 36×36, which are flattened into a $41{,}472$-dimensional feature vector.

The flattened features are passed to a two-layer fully connected classifier: a 128-dimensional hidden layer with sigmoid activation and dropout (0.5), followed by a final linear layer and sigmoid output producing the probability of the true streak class.

Compared to the original DeepStreaks architecture, we use padded convolutions to preserve spatial dimensions within each convolutional block, so that spatial resolution changes are controlled exclusively by pooling layers.

In addition to the lightweight VGG6 architecture, we evaluate deeper residual networks (ResNet18 [10] and ResNet50) to assess the impact of architectural depth. We also consider a metadata-only multilayer perceptron operating on eight detector-derived numerical features, namely the image-plane pixel coordinates and sky coordinates (Right Ascension and Declination) of the estimated streak endpoints $(x_{\text{start}}, y_{\text{start}}, x_{\text{end}}, y_{\text{end}}, \text{RA}_{\text{start}}, \text{Dec}_{\text{start}}, \text{RA}_{\text{end}}, \text{Dec}_{\text{end}})$, implemented as a small fully connected network with hidden dimensions (64, 32, 16). Finally, we evaluate a fusion architecture combining the VGG6 image encoder with this metadata branch through feature concatenation and a joint classification head trained end-to-end.

3.2 Training Setup

For each candidate, we rotate the image to align the detected segment horizontally and extract a cutout with fixed height (200 px) and variable width equal to the detected segment length plus a 100 px margin on each side. Cutouts are Z-scale normalized, converted to 8-bit grayscale, and

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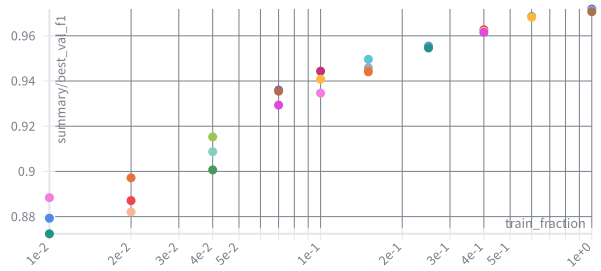


Figure 3: Validation F1-score vs training set fraction (log scale). Points denote different random seeds.

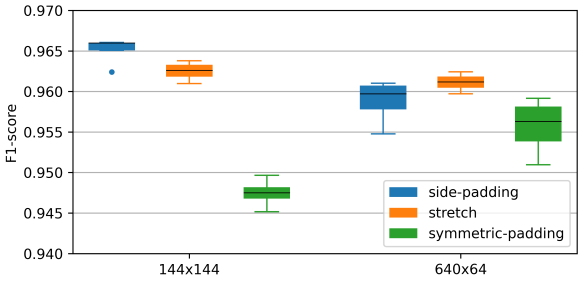


Figure 4: Validation F1-score for square and rectangular inputs under different resizing strategies (3 seeds per setting).

resized to 144×144 using horizontal padding. No data augmentation is applied.

We train the VGG6 architecture (Section 3.1) for binary classification using binary cross-entropy loss and the AdamW optimizer (learning rate 4.54×10^{-4}, weight decay 6.52×10^{-5}). Training is performed for 100 epochs with batch size 32 on detections from January to March 2023, and the model checkpoint with the highest validation F1-score is selected (random seed 84). Training on a single NVIDIA A100 (80 GB) requires approximately 1 h 12 min.

4 Model Design and Selection

We conduct a structured experimental study to evaluate the impact of data representation and model architecture on classification performance. Validation F1-score is used as the primary metric and results are averaged over multiple random seeds to account for training variance.

4.1 Data Representation

4.1.1 Training Set Size

We train the model on increasing fractions of the annotated dataset (1% to 100%). Validation F1-score improves steadily with additional data, but gains become progressively smaller at larger fractions, as shown in Figure 3. Considering the annotation effort required, we therefore use the full available training set and do not extend the labeled dataset further.

4.1.2 Rescaling and Padding

We compare square (144×144) and rectangular (640×64) inputs under three resizing strategies (Figure 4): `stretch` (direct resizing), `side-padding` (resize to target height with horizontal padding), and `symmetric-padding` (proportional resizing with padding on all sides).

The square format combined with side-padding achieves the highest validation F1-score. Although rectangular inputs better align with streak geometry, compressing the cutout into a compact square representation appears to concentrate the streak signal while reducing background noise, leading to improved generalization.

4.1.3 Cutout Context

We evaluate the effect of horizontal context around each streak. For rectangular inputs, larger margins improve performance but increase variance at extreme values. For square 144×144 inputs, larger margins do not yield consistent gains, likely due to compression into a fixed-size representation. We therefore retain a 100 px horizontal margin.

4.1.4 Image Bit Depth and Z-Scale Normalization

We evaluate 8-bit and 16-bit encodings with and without Z-scale normalization. Applying Z-scale normalization consistently improves validation performance, regardless of bit depth. Without contrast normalization, faint streaks remain poorly separated from background noise, leading to a clear degradation in F1-score. When Z-scale normalization is applied, 8-bit and 16-bit inputs achieve comparable results, indicating that contrast enhancement is more critical than raw photometric precision. We therefore retain 8-bit Z-scale normalized grayscale inputs in the final configuration.

4.1.5 Color Encoding

We evaluate multiple grayscale encodings and three-channel RGB inputs. Performance differences across colormaps are small, and RGB inputs do not provide consistent improvements over single-channel representations. We therefore retain a single-channel grayscale encoding for simplicity and computational efficiency.

4.1.6 Data Augmentations

We investigate common augmentation strategies including cropping, noise injection, synthetic CCD defects, and rotations. Most augmentations do not provide systematic improvements over the baseline. Rotations consistently degrade performance, as streaks are horizontally aligned by construction. Cropping also reduces performance by

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Model	F1	Params	Inference (s)
MetadataMLP	0.882	3.2K	7.7
ResNet18	0.961	11.2M	–
ResNet50	0.961	23.5M	–
VGG6	0.966	5.3M	29.5
VGG6+Meta	0.962	5.3M	34.9

Table 1: Model comparison (validation F1 averaged over 5 seeds; inference on 3,370 2023-12 detections using a Tesla V100 GPU).

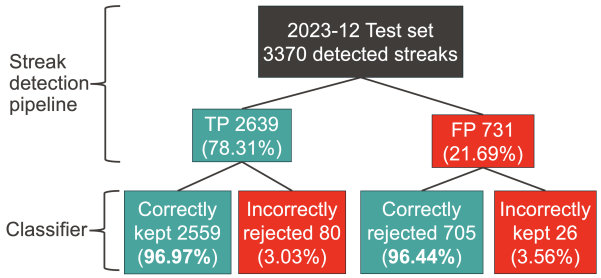


Figure 5: End-to-end performance on the 2023-12 test set.

removing contextual information. We therefore train the final model without data augmentation.

4.2 Architecture Comparison

To ensure fair comparison, we perform dedicated hyperparameter tuning for each architecture using random sweeps over learning rate and weight decay.

As shown in Table 1, neither the deeper residual networks nor metadata fusion improves over VGG6. The metadata-only model performs substantially worse than the image-based models. Permutation-based feature importance (measured by independently shuffling each feature and recording the decrease in validation F1-score) shows that the pixel coordinates of the endpoints have the greatest influence, while the sky coordinates contribute less. Manual inspection reveals a tendency to retain candidates spanning the whole CCD and reject those ending within it.

5 Results

We evaluate the final pipeline on an independent December 2023 test set comprising 3,370 streak candidates produced by the initial detection stage. This evaluation therefore measures the classifier's ability to reject false detections while retaining true ones among the detector outputs. Streaks missed by the detector are not included.

As shown in Figure 5, adding the classifier to the original detection pipeline reduces false positives (FP) by 96.44% while retaining 96.97% of true streaks.

The quantitative impact is detailed in Table 2, where the F1-score increases from 0.8783 to 0.9797, reflecting a substantial precision gain with minimal loss of true detections.

Metric	Raw	+ Classifier	Δ
Accuracy	0.7831	0.9685	+0.1854
Precision	0.7831	0.9899	+0.2068
F1-score	0.8783	0.9797	+0.1014

Table 2: End-to-end performance before and after adding the classifier (2023-12 test set).

FP class	Total	Rejected	Rejection rate
Fringes	356	352	98.88%
Diffraction	165	160	96.97%
Saturation	134	130	97.01%
Miscellaneous	76	63	82.89%

Table 3: Classifier performance across false positive categories on the December 2023 test set.

As detailed in Table 3, rejection rates exceed 96% for dominant artifact types (diffraction, saturation, fringes). The heterogeneous miscellaneous category has a lower rejection rate of 82.89% and accounts for 13 of the 26 false positives incorrectly retained after classification. Visual inspection of these 13 cases shows that 11 contain elongated, segmented structures resembling genuine streaks (Figure 6).

Manual inspection of the 80 true streaks incorrectly rejected by the classifier identifies six visual characteristics summarized in Table 4. Low contrast is by far the most frequent. Other recurring characteristics include crowded backgrounds, bright-source overlap, and broad or multiple traces. Representative examples are shown in Figure 7.

We further evaluate the classifier across the full range of decision thresholds. As shown in Figure 8, precision remains high over a broad range of recall values, resulting in an average precision of 0.993. This indicates that the reported performance is not tied to a specific decision threshold, and since threshold optimization on the validation set did not yield a meaningful improvement, we retain the standard threshold of 0.5.

6 Discussion

6.1 Interpretation of Results

The high rejection rates for the dominant artifact types show that the lightweight CNN effectively captures their recurring visual characteristics.

The remaining errors involve overlap between streak-like and artifact-like characteristics. Among rejected true streaks, low contrast predominates; crowded backgrounds, bright-source overlap, and broad or multiple traces also re-

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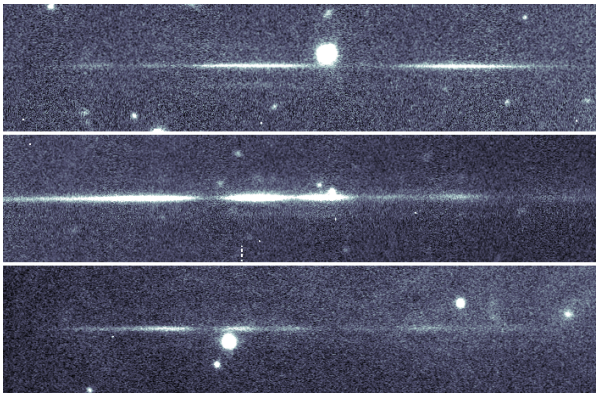


Figure 6: Examples of residual false positives from the miscellaneous category, showing elongated, segmented structures resembling streaks.

Visual characteristic	Count	Fraction
Low contrast	67	83.8%
Crowded stellar field	26	32.5%
Filter-fringe background	13	16.3%
Bright-source overlap	9	11.3%
Broad trace	16	20.0%
Multiple parallel traces	9	11.3%

Table 4: Visual characterization of the 80 true streaks incorrectly rejected by the classifier. Categories are non-exclusive.

cur. Conversely, residual false positives commonly contain segmented linear structures resembling genuine streaks.

The metadata-only model exploits geometric biases in the candidate set rather than artifact appearance. Since metadata fusion does not improve VGG6, endpoint coordinates provide little information beyond the cutout images. The absence of gains from deeper networks further suggests that the relevant visual patterns are already captured by the lightweight architecture.

6.2 Operational and Scientific Impact

From an operational perspective, the classifier enables large-scale archival processing. Classification inference for the 3,370 December 2023 candidates requires $\sim$ 30 seconds of GPU time. Scaling to the full 2023 archive remains within a few minutes.

By reducing false positives, the classifier drastically lowers manual inspection effort, allowing human validation to focus on a small subset of uncertain cases guided by confidence scores, while avoiding unnecessary downstream astrometric and photometric computations on spurious detections.

Beyond efficiency gains, reducing the false positive rate directly improves the scientific value of the catalog. A large-scale analysis of one year of OmegaCAM data [9] showed that roughly 20% of confirmed streaks cannot be correlated

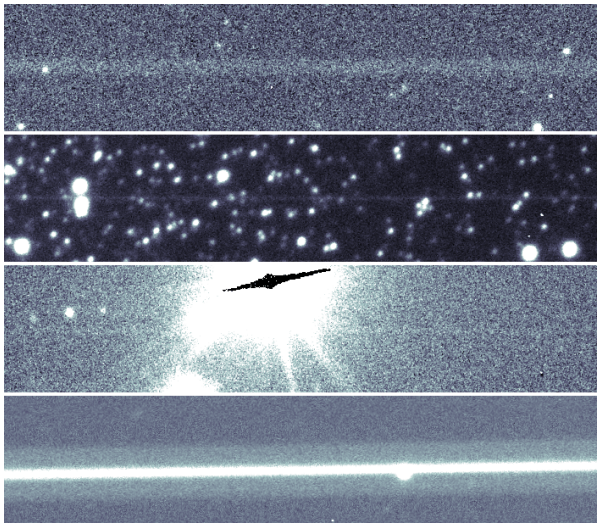


Figure 7: Examples of true streaks incorrectly rejected by the classifier, showing, from top to bottom, low contrast, a crowded stellar field, interference from a bright-source artifact, and a broad trace with parallel components.

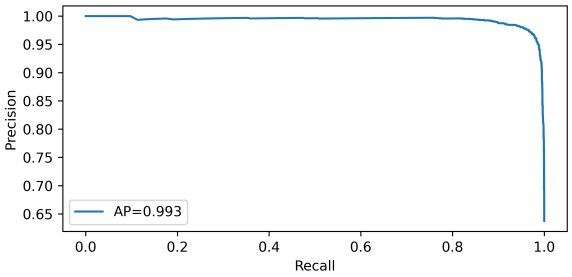


Figure 8: Precision-recall curve of the classifier on the 2023-12 test set. It achieves an average precision (AP) of 0.993.

with known catalogued objects. Many of these are faint detections that may correspond to small, uncatalogued debris. Ensuring a low false positive rate is therefore critical to make these detections reliable and suitable for further characterization.

6.3 Limitations and Future Work

Further gains are likely to come from increased data diversity rather than additional samples from the same period. Incorporating observations acquired under different observing conditions and across multiple seasons would improve robustness and allow a more rigorous assessment of generalization.

Extending the binary classifier to a multi-class artifact model is a natural next step. Explicitly distinguishing major false positive categories (e.g. fringes, diffraction spikes, saturation) would allow training specialized classifiers for each artifact type, or combining them in an ensemble.

Finally, adapting the approach to other telescope archives is another natural direction for future work.

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